Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Connected Humanoid Robots: Integration Strategy and Ecosystem: ISE-CHR			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Mechanical		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Robot Control and Intelligence Technologies	Software	System Integration
Concept of Proposed Technology	The proposed technology establishes a 6G-based reference architecture designed to enable the safe, stable, and real-time operation of humanoid and quadrupedal robots within advanced network environments. The concept focuses on the seamless integration of robotic systems into 5G/6G infrastructures to achieve the low latency and high reliability required for tele-operation and remote monitoring. Central to this approach is a centralized-distributed hybrid control architecture that supports multi-robot operations while addressing operational challenges such as high energy consumption and demanding uplink requirements. The framework incorporates secure end-to-end data pipelines for fleet learning, alongside cloud-based systems for remote maintenance and OTA updates. By validating these technologies within industrial testbeds and digital twin environments, the project provides a scalable and secure platform for the commercial deployment of robotic fleets in sectors			

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	such as smart manufacturing, security, and logistics.
Rationale for Joint R&D Project	The project leverages the strategic synergy between Korea's leading-edge humanoid robotics hardware and Germany's pioneering research in 5G/6G communication infrastructures. By integrating advanced Korean robotic platforms into established German research testbeds (e.g., 5G Industry Campus Europe at Fraunhofer IPT), the partnership bridges the gap between physical robot operation and network-centric control architectures. This collaboration allows for the joint definition of international standards and the validation of scalable, secure systems under diverse industrial conditions. Ultimately, this joint R&D effort facilitates a dual-market entry, combining European industrial excellence with Korean technological agility to establish a globally competitive reference for next-generation robotic fleets.
Key Words	1. 5G / 6G Enabled Robotic Operation 2. Reliable and automated teleoperation 3. Humanoid and quadruple Robotics for industrial processes 4. Centralized Humanoid Control

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	**Start of the Project**	Level (2)	**End of the Project**	Level (7)
R&D Objectives	The objective of this R&D project is to develop core technologies that enable safe, stable, and real-time operation of humanoid and quadrupedal robots over 5G/6G network environments. A. Ultra-low-latency, high-reliability tele-operation technologies B. Intelligent video transmission and edge processing to reduce communication costs C. End-to-end secure communication architecture resistant to hacking attacks D. Cloud-based OTA updates and remote maintenance systems E. Centralized–distributed hybrid control architecture for multi-robot operations F. Secure data pipelines for future imitation learning and fleet learning			
R&D Target Specifications	• System Synchronization: To achieve stable tele-operation of humanoid robots, the entire system loop (from the robot's sensors through the network to the backend) must remain under 30 ms to prevent physical instability or operator disorientation. • Operational Continuity: In industrial environments like smart factories, the system must ensure high continuity. This metric validates the robust integration of the robot into the moving cells of a 5G/6G infrastructure. • Scalability & Onboarding: Shifting from single prototypes to industrial fleets requires automated processes. A 120-second onboarding target demonstrates the architecture's "Plug-and-Play" capability, while the fleet density target proves the efficiency of the hybrid control model. • Maintenance & Security: As the robots rely on cloud-based pipelines for fleet learning and updates, a high OTA success rate is critical to minimize downtime. Furthermore, the Security Response Time ensures that the integrated system can react to cyber threats faster than a human operator could intervene.			
R&D Plan	A. Year 1 – Foundation & Architecture i. Integration with Fraunhofer 5G testbed ii. Standardization of robot communication interfaces iii. Baseline tele-operation and video streaming pipeline iv. Security threat analysis and requirement definition B. Year 2 – Core Technology Development v. Low-latency control protocols and network slicing vi. Edge-based video summarization and event-driven streaming vii. Security modules: authentication, intrusion detection, access control viii. OTA update and remote diagnostics system C. Year 3 – Validation & Demonstration ix. Multi-robot simultaneous operation trials x. Failure and cyberattack scenario testing xi. Validation of cloud-based data collection for imitation learning xii. Final system architecture for commercialization			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Integration of humanoid and quadrupedal robots into 5G / 6G network • Provision of massive sensor data support • Securing end-to-end communication with guaranteed QoS • Development of according cyber security strategy • Development of centralized control together with robotic SME • Providing different robotic usecases (e.g. manufacturing, logistics, rescue) • Define commercialization-oriented system requirements
Expected R&R of Korean SME Partner	• Provide and operate humanoid and quadrupedal robots • Supply real-world operational data and failure cases • Design and execute testbed demonstration scenarios together with Research Institute • Define commercialization-oriented system requirements

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Reference architecture for 5G/6G-based robot tele-operation • Cost-efficient integrated control and video communication system • Secure robot operation model resilient to cyber threats • Scalable robot operation platform supporting OTA updates and cloud learning • Foundation for global deployment through EU–Korea collaboration The developed technologies can be commercialized through Korean telecom operators' Private 5G/MEC offerings and scaled across shipbuilding, construction, smart factories, and residential security in Korea, with a clear pathway to expand into the U.S. security market via fleet-based remote patrol and command-center operations.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization